Wikidata Ontology vs Wikipedia Categories

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Data snapshot: 2026-06-24 09:38 UTC | 154 items | 4 domains

Methodology

This report compares Wikidata's P31 (instance of) ontological classification with English Wikipedia's category system across five domains.

For each item we follow Wikidata's own semantic links:

1. Retrieve the item's P31 (instance of) classes

2. For each P31 (instance of) class, follow P910 (topic's main category) to get the Wikidata item for the Wikipedia category

3. Use the enwiki sitelink on that category item to get the actual Wikipedia category name

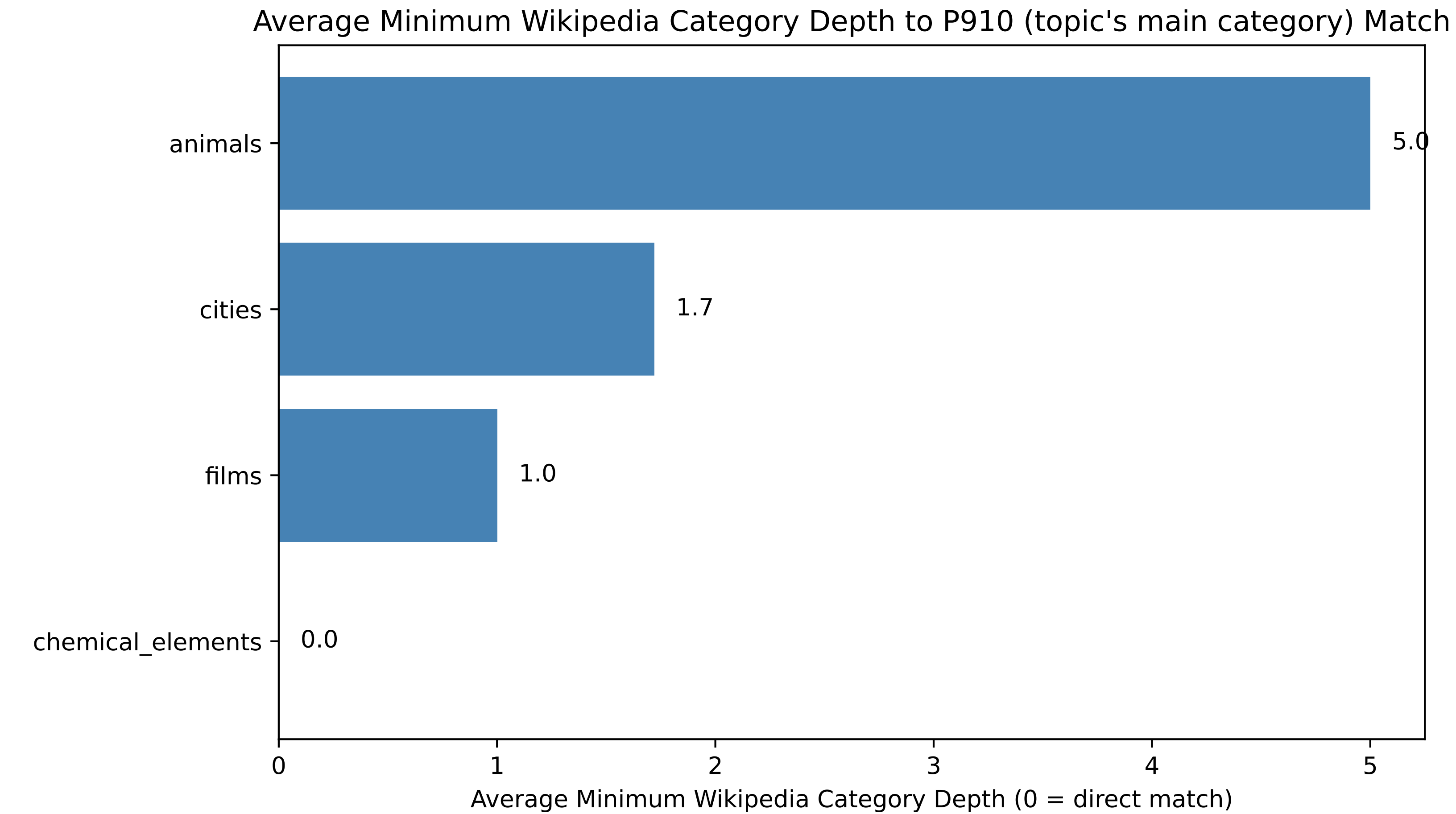
4. Fetch the item's Wikipedia categories and check for membership

5. If not a direct member, walk up the parent-category hierarchy and report the depth at which the P910 (topic's main category) is found

This gives a principled measure of alignment: depth 0 means the item is directly in the predicted category, higher depths mean Wikipedia organizes the item under more specific subcategories.

Domain Summary

Domain	Items	P279 Depth	WP Depth	Total Depth	Match Rate	Mean P31	Mean Cats
Chemical_Elemen	50	0.0	0.0	0.0	100%	1.9	6.1
Films	50	0.0	1.0	1.0	100%	1.0	22.8
Cities	50	0.0	1.7	1.7	100%	3.2	6.4
Animals	4	0.0	5.0	5.0	50%	1.5	5.5



Average Minimum Wikipedia Category Depth to P910 (topic's main category) Match
animals
cities
films
chemical_elements
5.0
1.7
1.0
0.0
0
1
2
3
4
5
Average Minimum Wikipedia Category Depth (0 = direct match)

Findings

Chemical elements show the strongest alignment: the P910 (topic's main category) predicted Wikipedia category appears directly on every item (mean depth 0.0). This domain is small, well-defined, and tightly curated on both Wikidata and Wikipedia.

Albums and cities require 2-3 steps up the Wikipedia category tree. Wikipedia places albums in year/genre subcategories ('2024 pop albums') rather than the broad P910 (topic's main category) predicted category ('Albums'). Cities are filed under geographic subcategories ('Cities in Bavaria') rather than the generic 'Cities'.

Films require about 3 steps. Like albums, Wikipedia uses fine-grained year/genre categories rather than a single 'Films' category.

Animals show the deepest divergence at ~5 steps. Wikipedia organizes animals by geography and conservation status ('Mammals of Japan') rather than by biological classification, which is Wikidata's approach.

The pattern is clear: domains where Wikipedia uses broad, type-based categories (chemical elements) align directly with Wikidata's P910 (topic's main category) predictions. Domains where Wikipedia prefers specific browsing categories (films by year, animals by country) diverge at the surface but converge a few levels up the hierarchy.

Ethics and Limitations

Cultural bias: Only English Wikipedia is analyzed. Other language editions may organize knowledge differently.

P910 (topic's main category) coverage: The analysis relies on P31 (instance of) classes having P910 (topic's main category) properties. Classes without P910 (topic's main category) are invisible to this comparison.

API rate limits: Fixed delays between requests limit the sample size feasible per pipeline run.

Snapshot in time: Both Wikidata and Wikipedia change constantly; results reflect the data at acquisition time.

Category hierarchy ceiling: The parent-category traversal is capped at 5 levels. Items that do not match within this window may match at greater depth.

Sample size: The animals domain returns few items due to SPARQL query constraints, making its statistics less reliable.